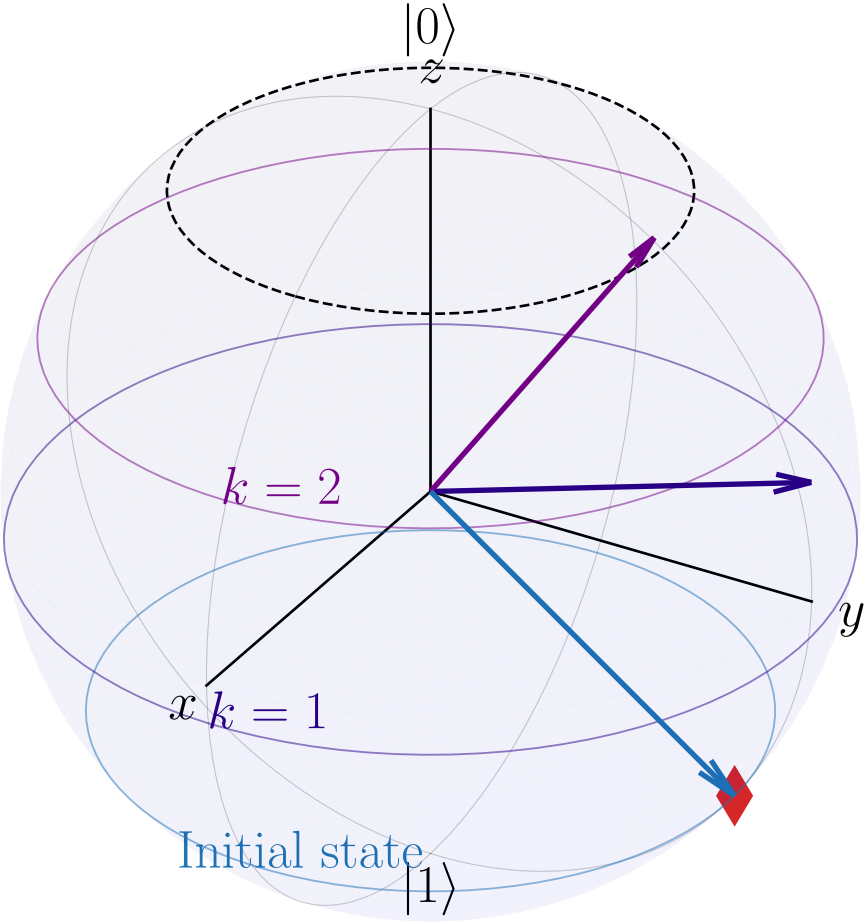

$|0\rangle$
z
$k = 2$
y
x
$k = 1$
Initial state
$|1\rangle$